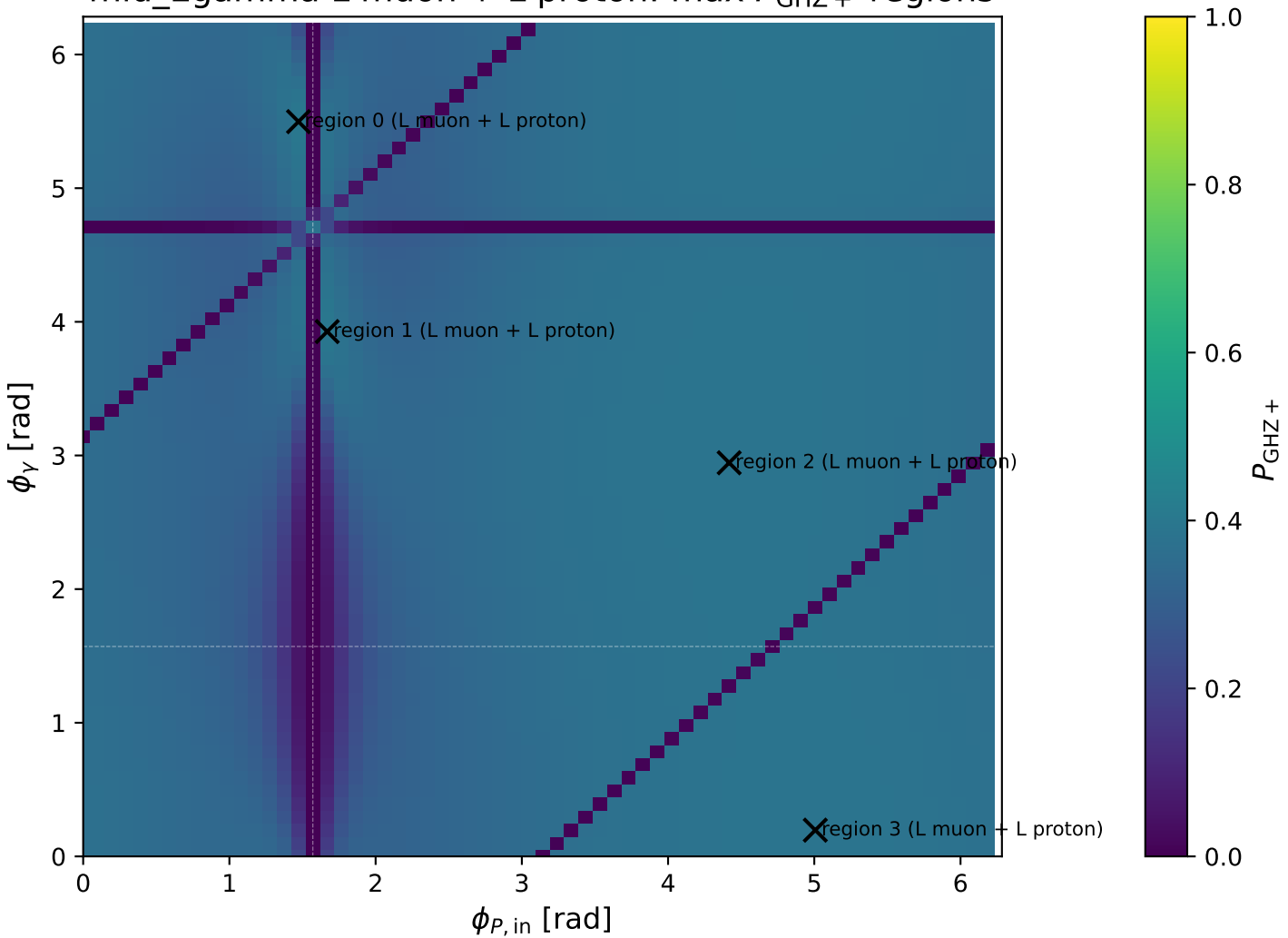
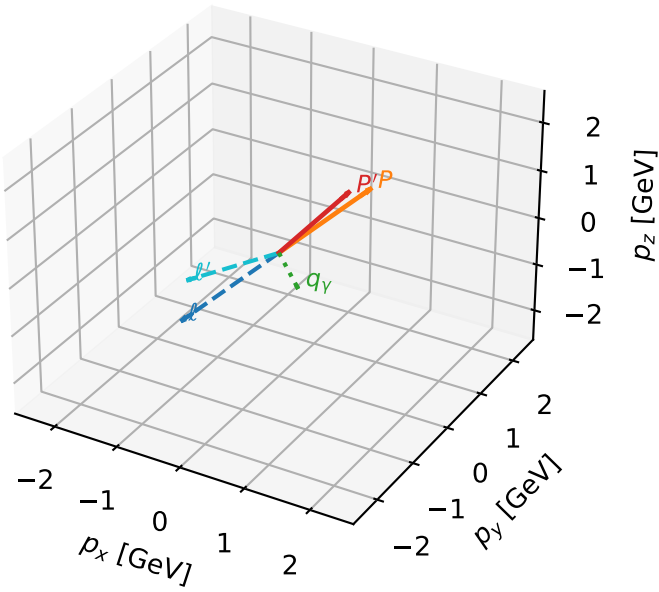


mid_Egamma L muon + L proton: max P_{GHZ} + regions



L muon + L proton characteristic max P_{GHZ} + configuration

3D momenta

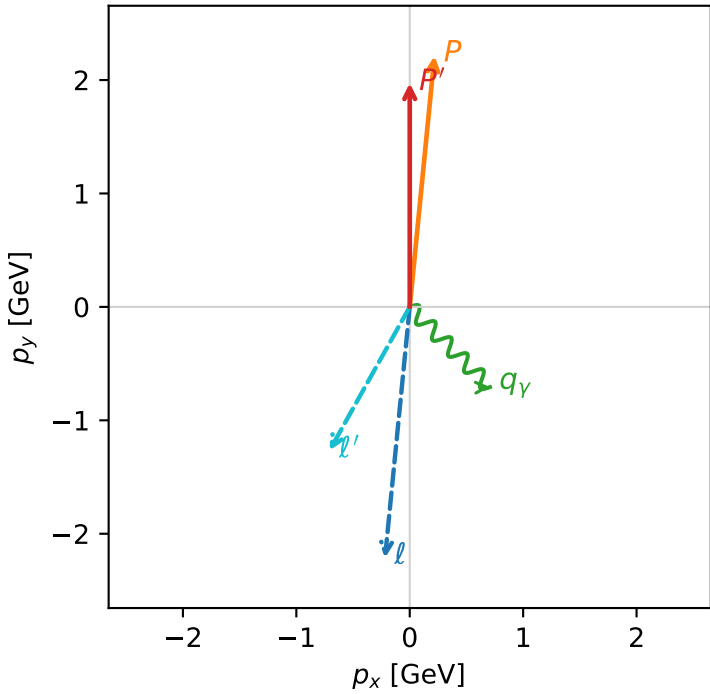


ghz_purity_mid_egamma_ll_region_0 $P_{\{\text{rm GHZ}\}}=0.395808$
region: mid_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

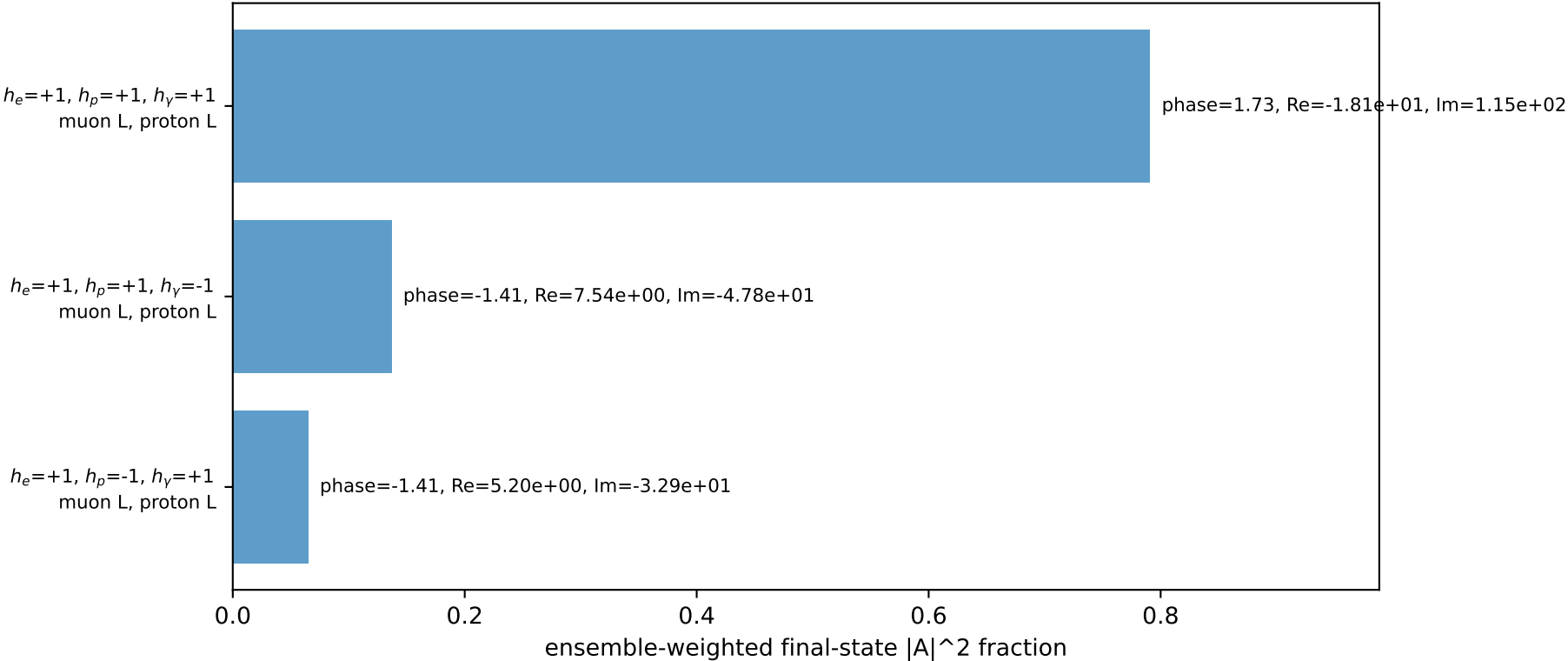
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.97303$
 $\theta_{\text{in}}=1.57079$, $\phi_l=4.61421$, $\phi_p=1.47262$
 $E_Y=1$, $\phi_Y=5.49779$
 $Q^2=0.539544$, $x_B=0.0700786$, $t=-0.052686$, $W^2=8.03943$, $y=0.373294$
 $\theta(l', \gamma)=74.1864$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.2179$ $p_y= -2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= 2.2124$ $p_z= 0.0000$
 l' $E= 1.4539$ $p_x= -0.7071$ $p_y= -1.2659$ $p_z= 0.0000$
 P' $E= 2.1847$ $p_x= 0.0000$ $p_y= 1.9730$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.7071$ $p_y= -0.7071$ $p_z= 0.0000$

Transverse plane

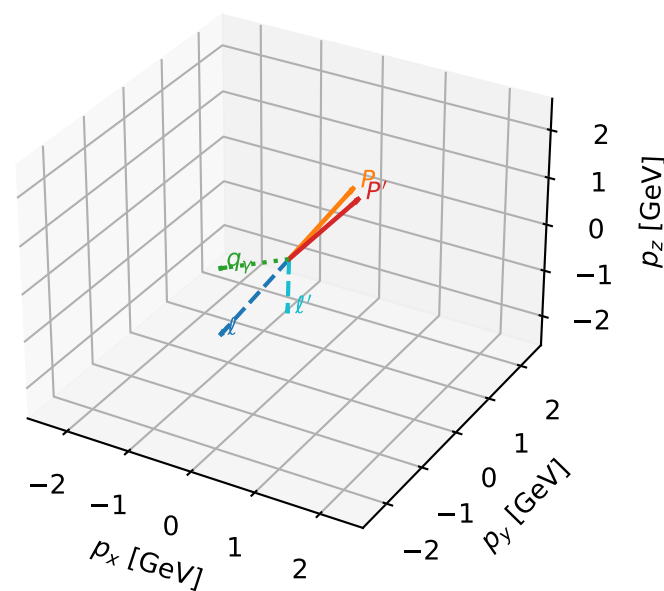


Leading initial-ensemble/final-state components (N=3, retained=99.2%)



L muon + L proton characteristic max P_{GHZ} + configuration

3D momenta

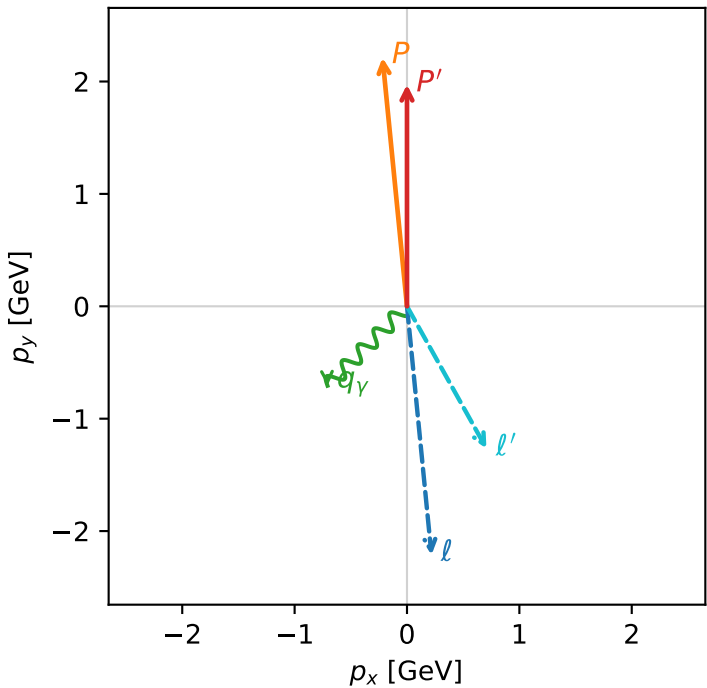


ghz_purity_mid_egamma_ll_region_1 $P_{\{\text{rm GHZ}\}}=0.395808$
region: mid_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

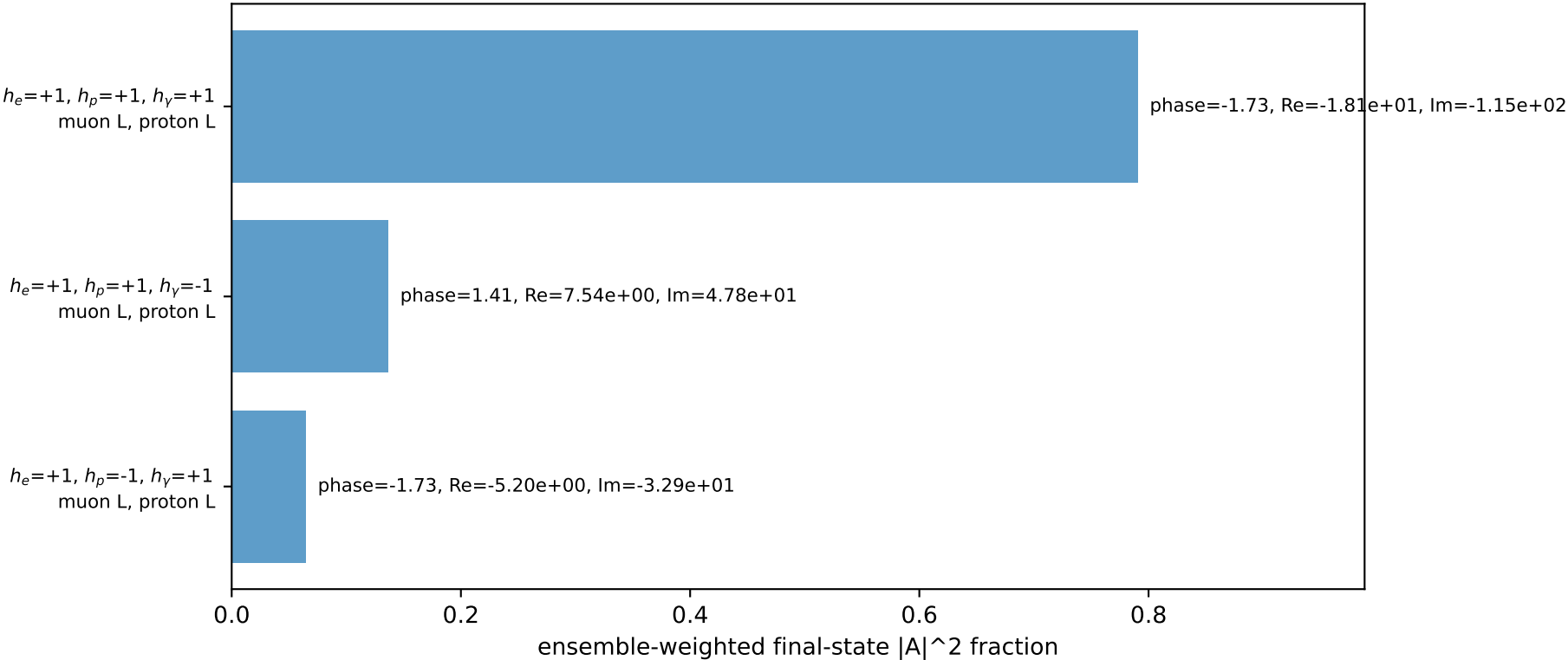
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.97303$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=4.81056$, $\phi_P=1.66897$
 $E_Y=1$, $\phi_Y=3.92699$
 $Q^2=0.539544$, $x_B=0.0700786$, $t=-0.052686$, $W^2=8.03943$, $y=0.373294$
 $\theta(\ell', \gamma)=74.1864$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= 0.0000$
 ℓ' $E= 1.4539$ $p_x= 0.7071$ $p_y= -1.2659$ $p_z= 0.0000$
 P' $E= 2.1847$ $p_x= 0.0000$ $p_y= 1.9730$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.7071$ $p_y= -0.7071$ $p_z= 0.0000$

Transverse plane

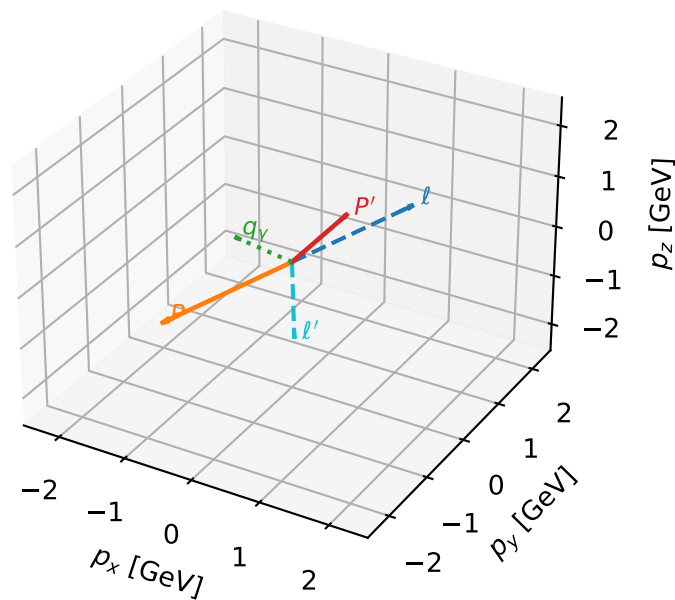


Leading initial-ensemble/final-state components (N=3, retained=99.2%)



L muon + L proton characteristic max P_{GHZ} + configuration

3D momenta

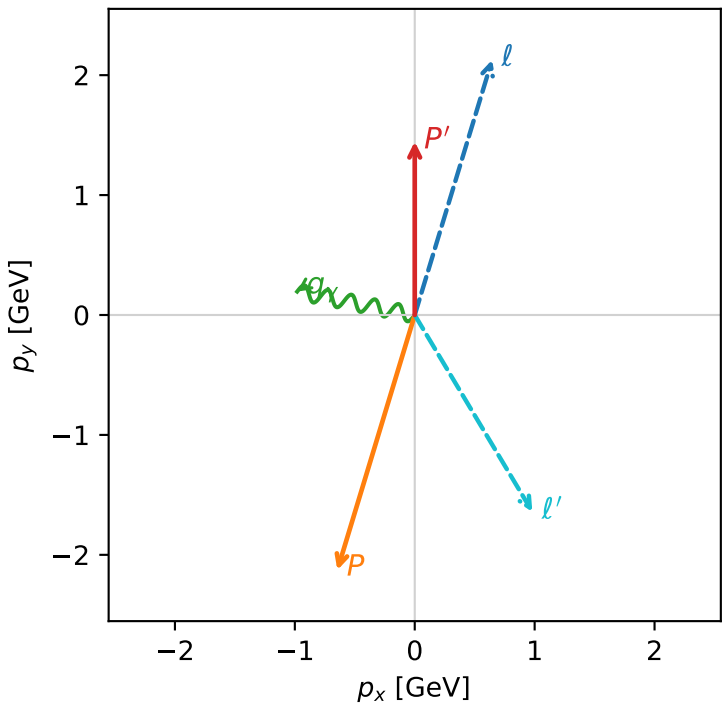


ghz_purity_mid_egamma_ll_region_2 $P_{\text{GHZ}}=0.385481$
region: mid_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

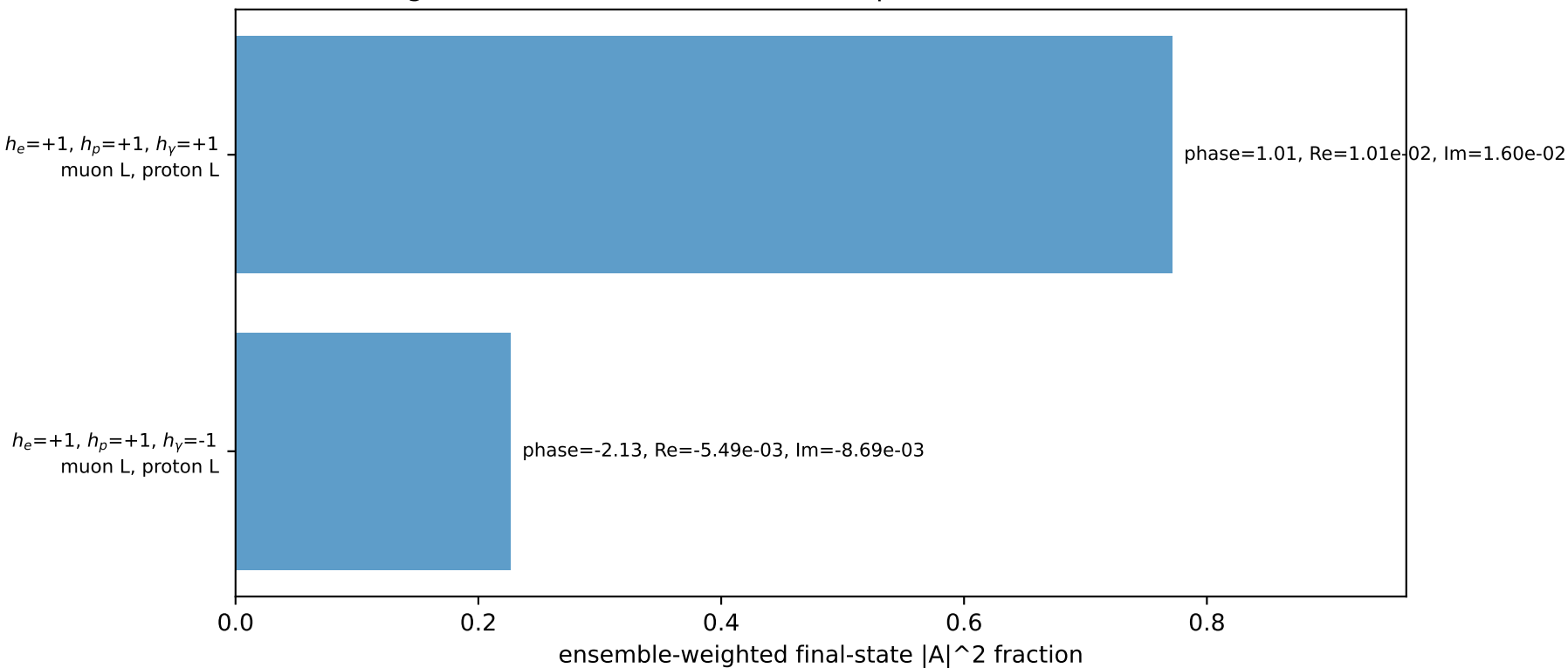
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.4461$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.27627$, $\phi_p=4.41786$
 $E_\gamma=1$, $\phi_\gamma=2.94524$
 $Q^2=14.2181$, $x_B=0.831409$, $t=-12.7112$, $W^2=3.76295$, $y=0.829157$
 $\theta(l', \gamma)=132.113$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.6453$ $p_y= 2.1274$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.6453$ $p_y= -2.1274$ $p_z= 0.0000$
 l' $E= 1.9148$ $p_x= 0.9808$ $p_y= -1.6412$ $p_z= 0.0000$
 P' $E= 1.7237$ $p_x= 0.0000$ $p_y= 1.4461$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.9808$ $p_y= 0.1951$ $p_z= 0.0000$

Transverse plane

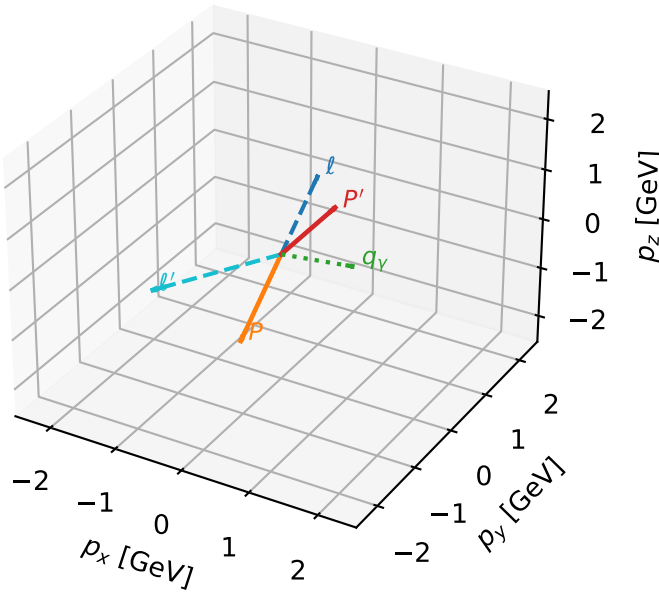


Leading initial-ensemble/final-state components (N=2, retained=99.8%)



L muon + L proton characteristic max P_{GHZ} + configuration

3D momenta

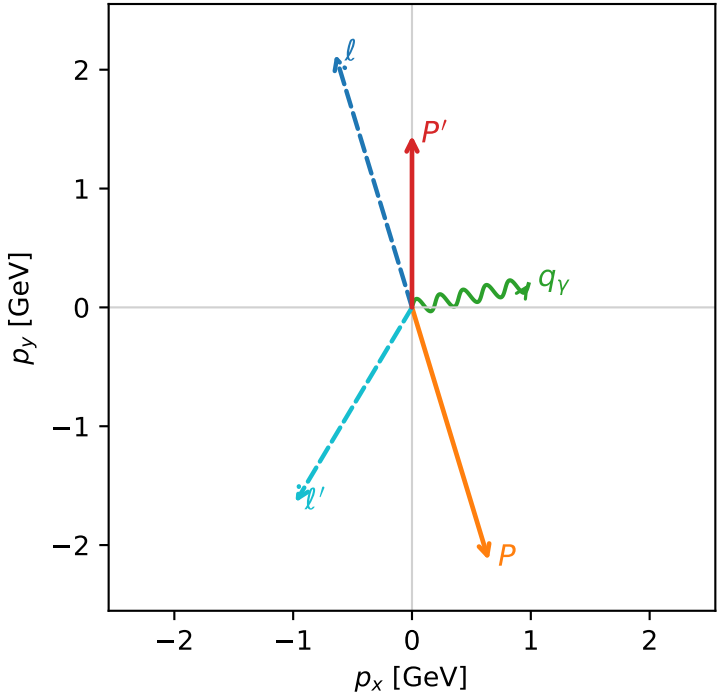


ghz_purity_mid_egamma_ll_region_3 $P_{\text{GHZ}}=0.385481$
region: mid_Egamma_LL_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.4461$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.86532$, $\phi_P=5.00691$
 $E_\gamma=1$, $\phi_\gamma=0.19635$
 $Q^2=14.2181$, $x_B=0.831409$, $t=-12.7112$, $W^2=3.76295$, $y=0.829157$
 $\theta(l', \gamma)=132.113$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.6453$ $p_y= 2.1274$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.6453$ $p_y= -2.1274$ $p_z= 0.0000$
 l' $E= 1.9148$ $p_x= -0.9808$ $p_y= -1.6412$ $p_z= 0.0000$
 P' $E= 1.7237$ $p_x= 0.0000$ $p_y= 1.4461$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9808$ $p_y= 0.1951$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.8%)

